■ 6-MONTH DATA SCIENCE ROADMAP

Python → Data Analysis → Visualization → ML → Real Projects

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6	24	4–5 hrs	1
Months	Weeks	Daily	Project/Week

■ DAILY STUDY PLAN (4–5 hours)

2 hrs	2 hrs	1 hr
■ Learning	■ Coding	■ Analysis Work

■ RULES TO FOLLOW — NO SHORTCUTS

■	Don't just study theory — code karo aur analyze karo, har roz
■	Don't copy-paste code — understand every single line you write
■	Work on real datasets every week — theory alone won't get you a job
■	Build projects combining all skills — push everything to GitHub

■■ TECH STACK YOU'LL USE

■ Python	■ NumPy	■ Pandas	■ Matplotlib	■ Seaborn	■ Scikit-learn

■ 6-MONTH JOURNEY

M1	M2	M3	M4	M5	M6
Python + Data Handling	Visualization	Statistics + EDA	ML Basics	Advanced ML	Projects + Deploy

■ MONTH 1 — Python + Data Handling

Month 1	Python + Data Handling
Week 1	■ Python Basics — variables, data types, operators ■ Control Flow — if/elif/else, loops (for, while) ■ Functions — defining, parameters, return statements ■ *...and more*
Week 2	■ Data Structures — lists, tuples, dictionaries, sets ■ List comprehensions aur slicing techniques ■ String manipulation aur built-in methods ■ *...and more*
Week 3	■ NumPy — arrays, shapes, operations ■ Broadcasting, indexing, reshaping arrays ■ Mathematical operations on arrays — vectorization ■ *...and more*
Week 4	■ Pandas — Series, DataFrames, basic operations ■ Reading CSV files, exploring data with head/info/describe ■ Data Cleaning — handling missing values, duplicates ■ *...and more*
■ BUILD	■ CSV Data Analysis ■ Data Cleaning Project ■ Basic Exploratory Analysis

■ MONTH 2 — Data Visualization

Month 2	Data Visualization
Week 1	■ Matplotlib Basics — figures, subplots, axes ■ Line plots, scatter plots, bar charts ■ Customization — colors, labels, titles, legends ■ *...and more*
Week 2	■ Histogram aur distribution plots ■ Box plots for outlier detection ■ Combining multiple plots effectively in one figure ■ *...and more*
Week 3	■ Seaborn — statistical visualization library ■ Heatmaps for correlation matrices ■ Categorical plots — count, bar, violin, strip ■ *...and more*
Week 4	■ Advanced visualization — dashboard style layouts ■ Telling stories with data visuals — design principles ■ Practice week — visualize multiple real datasets ■ *...and more*
■ BUILD	■ Sales Data Visualization ■ Dataset Analysis Report ■ Interactive Charts

■ MONTH 3 — Statistics + EDA

Month 3	Statistics + Exploratory Data Analysis
Week 1	■ Descriptive Statistics — mean, median, mode, variance, std dev ■ Quantiles aur percentiles — understanding spread ■ Distribution analysis — skewness, kurtosis ■ *...and more*
Week 2	■ Probability Basics — basic rules, distributions ■ Normal distribution aur z-scores ■ Hypothesis Testing — p-values, confidence intervals ■ *...and more*
Week 3	■ Correlation aur covariance — measuring relationships ■ Feature relationships — correlation matrix, heatmaps ■ Finding patterns aur anomalies in data ■ *...and more*
Week 4	■ EDA (Exploratory Data Analysis) — full end-to-end workflow ■ Data profiling aur quality checks ■ Write complete EDA report on a real dataset ■ *...and more*
■ BUILD	■ Statistical Analysis Report ■ Complete EDA Project ■ Data Quality Assessment

■ MONTH 4 — Machine Learning Basics

Month 4	Machine Learning Fundamentals
Week 1	■ ML Fundamentals — supervised vs unsupervised learning ■ Train-test split, overfitting, underfitting ■ Evaluation Metrics — accuracy, precision, recall, F1 ■ *...and more*
Week 2	■ Linear Regression — predicting continuous values ■ Logistic Regression — binary classification problems ■ Implementation using Scikit-learn — fit, predict, score ■ *...and more*
Week 3	■ KNN (K-Nearest Neighbors) — implementation aur tuning ■ Decision Trees — how they work, visualization ■ Feature importance aur tree pruning techniques ■ *...and more*
Week 4	■ Complete ML project — house price prediction end-to-end ■ Full workflow — data cleaning → training → evaluation ■ Model comparison aur selection — which algo to use when ■ *...and more*
■ BUILD	■ House Price Prediction ■ Iris Classification ■ Spam Classifier

■ MONTH 5 — Advanced Machine Learning

Month 5	Advanced ML + Model Improvement
Week 1	■ Feature Engineering — creating new meaningful features ■ Feature Selection — dropping irrelevant features ■ Scaling aur normalization — StandardScaler, MinMaxScaler ■ *...and more*
Week 2	■ Random Forest — ensemble methods, feature importance ■ Gradient Boosting — XGBoost, LightGBM basics ■ Model comparison aur stacking techniques ■ *...and more*
Week 3	■ Cross-validation — k-fold, stratified validation ■ Hyperparameter Tuning — GridSearch, RandomSearch ■ Advanced Metrics — ROC-AUC, precision-recall curves ■ *...and more*
Week 4	■ Customer churn prediction — complete real-world project ■ Real-world dataset handling — messy, imbalanced data ■ Model performance optimization aur final tuning ■ *...and more*
■ BUILD	■ Customer Churn Prediction ■ Credit Risk Assessment ■ Sales Forecasting

■ MONTH 6 — Projects + Deployment

Month 6	Real-World Projects + Deployment
Week 1	■ Model Serialization — saving aur loading models (pickle/joblib) ■ Flask/FastAPI — creating REST API endpoints for ML models ■ Testing your API with Postman aur curl ■ *...and more*
Week 2	■ Building web applications with ML models backend ■ Frontend integration with backend — connecting UI to API ■ Testing aur debugging ML applications end-to-end ■ *...and more*
Week 3	■ Dashboard creation — data visualization dashboards ■ Business intelligence with data science insights ■ Documenting projects aur writing clear README files ■ *...and more*
Week 4	■ Build 2–3 complete projects end-to-end ■ Real-world dataset projects — Kaggle competitions ■ Portfolio building aur deployment — GitHub + Render/HuggingFace ■ *...and more*
■ BUILD	■ Recommendation System ■ Stock Prediction Dashboard ■ Customer Analytics Portal

■ PROJECT MILESTONE TRACKER

Month	Key Deliverables
Month 1	CSV Data Analysis, Data Cleaning Projects
Month 2	Sales Analysis, Dataset Visualization Report
Month 3	EDA Report, Statistical Analysis Project
Month 4	House Price Prediction, Iris Classification
Month 5	Churn Prediction, Credit Risk Assessment
Month 6	Recommendation System, Stock Prediction Dashboard

■ RESOURCES — Everything You Need to Become a Data Scientist

■ OFFICIAL DOCUMENTATION

■ **Syntax Error**	syntaxerrorr.com	Data science notes, tutorials, DSA + placement resources — our website
■ **Pandas Docs**	pandas.pydata.org	Complete data manipulation and analysis reference — must bookmark
■ **Seaborn Docs**	seaborn.pydata.org	Statistical data visualization library — full API reference with examples

■■ ONLINE COURSES & TUTORIALS

■ BEST FOR BEGINNERS	**Krish Naik — Data Science** by Krish Naik youtube.com/@krishnaik06 Best for beginners — clear explanation + real datasets + code along every video. Covers Python, ML, DL, and deployment end-to-end.
■ HANDS-ON PRACTICE	**Kaggle Learn** by Kaggle kaggle.com/learn Free micro-courses on Python, data analysis, ML with hands-on practice notebooks. Best for practical skill building with real competitions.
■ STATS + ML CONCEPTS	**StatQuest with Josh Starmer** by Josh Starmer youtube.com/c/joshstarmer Master statistics and ML concepts with clear visual explanations. The best channel to deeply understand WHY algorithms work the way they do.

■■ DATASETS FOR PRACTICE

■ **Kaggle**	kaggle.com/datasets	Thousands of datasets aur competitions — the #1 place to practice
■ **UCI ML Repository**	archive.ics.uci.edu	Classic datasets for ML analysis — iris, wine, breast cancer and more
■ **Google Dataset Search**	datasetsearch.research.google.com	Find any dataset worldwide — Google-powered dataset discovery

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"Data is the new oil, but insights are the new gold. Learn to extract value from data — one dataset at a time. ■"